Coffee Shop Sales Analysis

Identifying revenue trends & customer behaviour

1. Introduction

What This Dataset Is About

This dataset contains transactional sales data from a coffee shop, capturing details of daily operations such as orders, products, quantities, and revenue. It provides insights into how different items perform, when sales peak, and how customers purchase across various time periods.

What Problem We Are Solving

The objective of this project is to transform raw transactional data into meaningful insights by analysing sales performance, product demand, and time-based trends, enabling the business to make informed decisions that improve profitability and operational efficiency.

The goal is to generate actionable insights that help optimize sales, and improve overall business decision-making.

2. Objectives / Analysis Questions

The following core questions drive this analysis:

- What is the overall sales performance in terms of total revenue, number of transactions, and average order value?
- Which products and categories contribute the most to revenue and sales volume?
- At what times of the day do peak sales occur, and how do sales vary across different time periods?
- What are the top-selling items, and which products show consistently low demand?
- How do customer purchasing patterns impact overall sales performance (high-frequency vs high-value orders)?
- How concentrated is revenue across products (dependency on top-performing items)?

3. Dataset Information

Source

Kaggle – csv form

Key Fields

- Order Date
- Revenue
- Order Time
- Product Name

4. Tools used

Excel – Data cleaning, analysis and dashboard creation.
SQL – Data querying, transformation and extraction.

5. Key KPIs

- Total Revenue – $~112K
- Average Order Value (AOV) – $31.64

6. Dashboard Overview

The dashboard provides a structured view of e-commerce performance:

- **Top KPIs** – Quick summary of revenue and average order value.
- **Revenue Analysis** – Product-wise contribution to total revenue.
- **Time Series Analysis** – Comparison of units sold across the day.
- **Yearly Analysis** – Analyses the revenue trend over the months.
- **Month Filter** – Interactive selection to analyse the month-wise performance.
- **Product Filter** – Interactive selection to analyse individual product performance.

7. Key Insights

- Coffee-based beverages contribute the highest share of total revenue, making them the core business drivers.
- Sales exhibit strong time-based patterns, with peak demand during specific hours (especially morning rush and evening).
- A small group of top-selling products generates a significant portion of overall sales
- High-frequency, low-value transactions dominate order volume, while fewer high-value orders contribute more to revenue.
- Certain products show consistently low demand, indicating opportunities for menu optimization.
- Revenue is concentrated among a few key items, highlighting dependency on top performers.

8. Business Recommendations

- Introduce some cold beverages and seasonal drinks to stabilize the falling revenue in the months of summer.
- Increase staffing and inventory during the peak hours (9-11 AM & 3-5 PM) to reduce waiting time.
- Leverage the digital only payment method by bringing some loyalty points/rewards through digital wallet.
- Also can experiment with extended evening hours on selected days to capture the night demand.

9. Conclusion

The analysis highlights key patterns in sales, product performance, and peak hours, enabling more informed and efficient business decisions. Leveraging these insights can help optimize operations, improve revenue, and enhance overall customer experience.